

Welcome back!



Topics

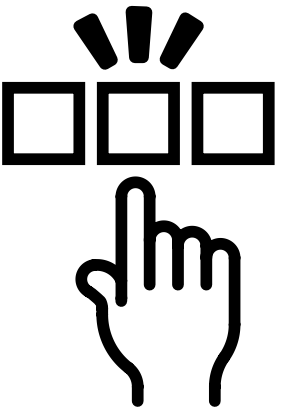
Project Introduction

Upcoming Activities:

- Watching YouTube videos
- Project introduction and overview

**Welcome to the
Personalized Movie
Recommendation
System Project!**

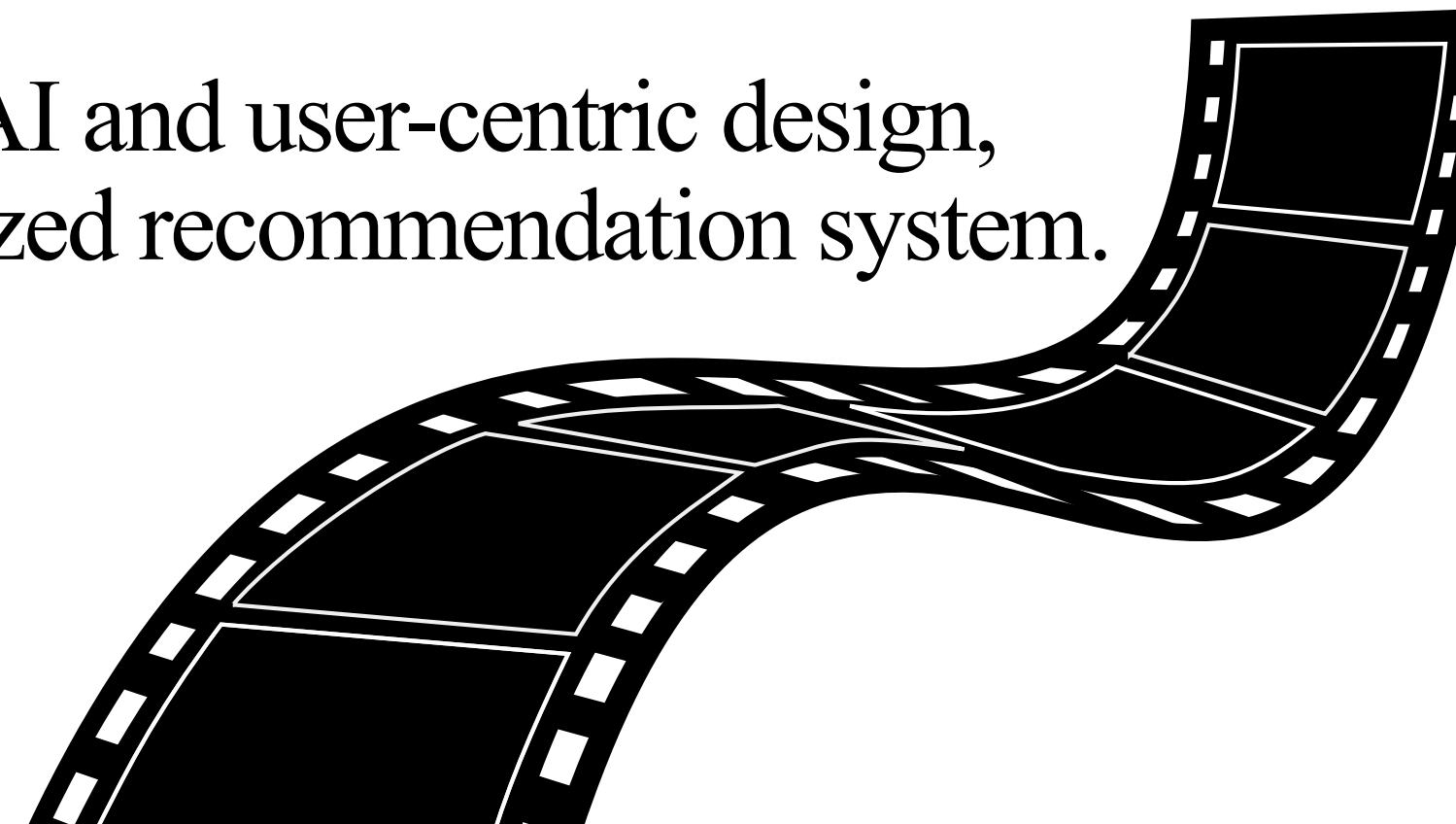
Overview



In this individual project, you will build a movie recommendation system that tailors suggestions based on user preferences and behavior.

Over the next eight weeks, you will work independently to collect user data, develop and train an AI model, design and integrate a user interface, and refine the system based on user feedback.

This project aims to enhance your technical skills in AI and user-centric design, providing practical experience in creating a personalized recommendation system.



Weekly Breakdown

Week 1: Project Introduction and Planning

Week 2: Understanding User Needs and Data Collection

Week 3: Setting Up AI Tools and Environment

Week 4: Data Preprocessing and Exploration

Week 5: Building the Recommendation Model

Week 6: Integrating the Model with a User Interface

Week 7: Testing and Refinement

Week 8: Final Presentation and Documentation

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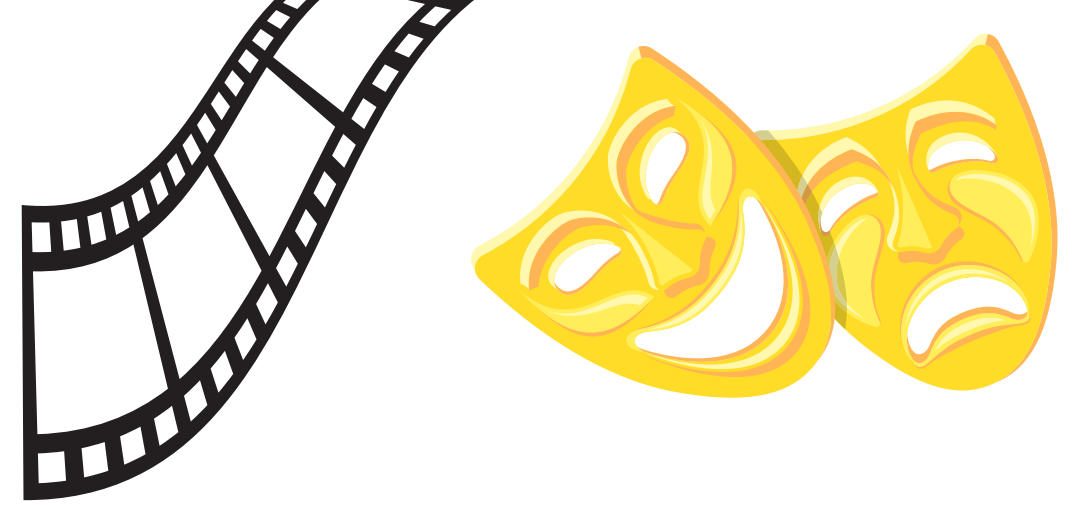
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Week 8: Final Presentation and Documentation

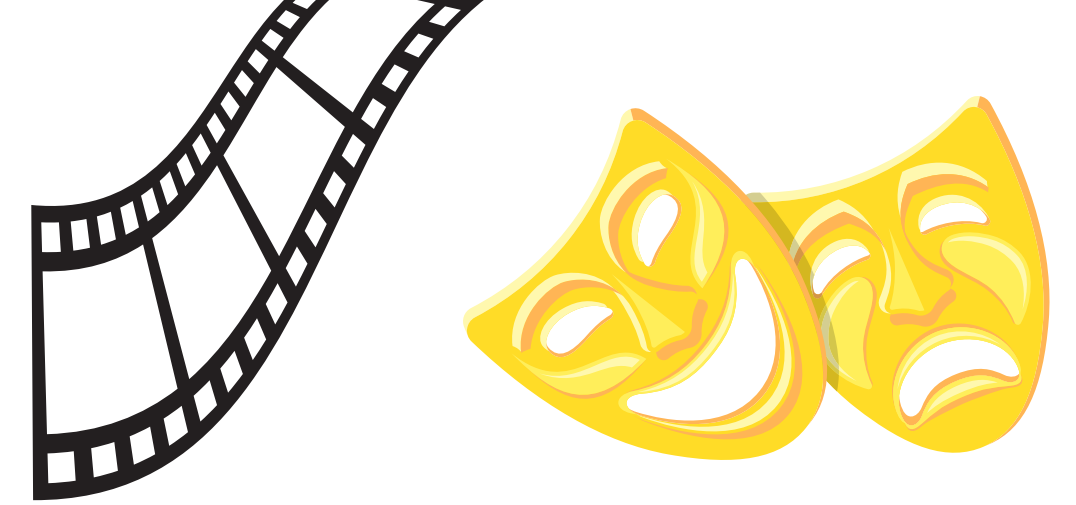
THIS is Week 1

Project Requirements



- 1. Project Planning:** (a) Develop a comprehensive project plan. (b) Include a timeline with milestones and deliverables. (c) Outline specific tasks to be completed each week.
- 2. User Needs Assessment and Data Collection:** (a) Conduct surveys or interviews to understand movie preferences and viewing behavior. (b) Collect relevant data for training the AI model (e.g., user ratings, movie genres, viewing history). (c) Ensure data privacy and ethical handling of user information.
- 3. AI Tool Setup:** (a) Select an appropriate AI tools and platforms (e.g., TensorFlow, PyTorch, AWS, Google Cloud). (b) Set up the development environment (e.g., Jupyter notebooks, IDEs). (c) Install necessary libraries and familiarize yourself with the tools through tutorials and practice exercises.

Project Requirements

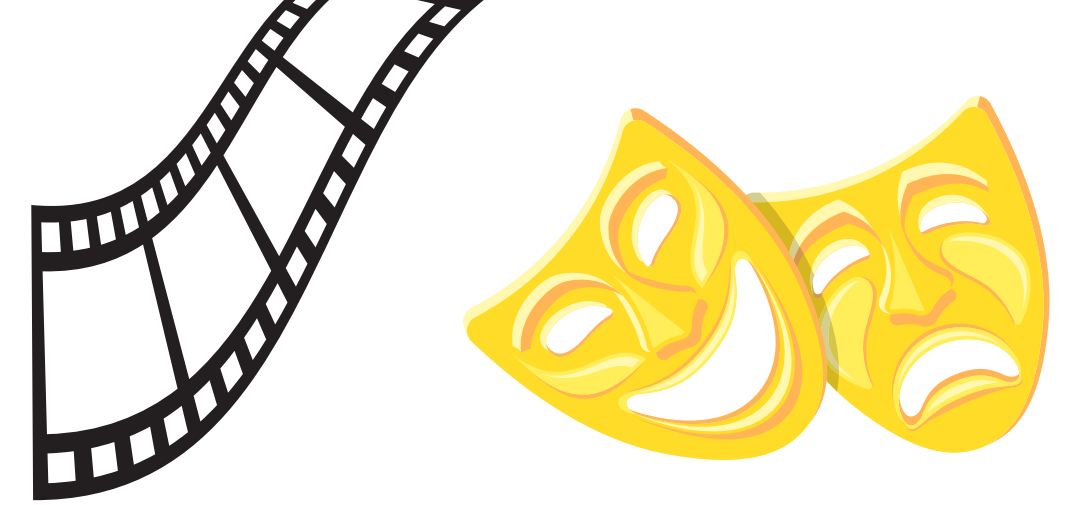


4. Data Preprocessing and Exploration: (a) Clean and preprocess the collected data (e.g., handle missing values, normalize data). (b) Explore the data using visualization tools to identify patterns and insights.

5. AI Model Development: (a) Choose an appropriate algorithm for the recommendation system (e.g., collaborative filtering, content-based filtering). (b) Build and train the AI model using the preprocessed data. (c) Evaluate the model's performance using metrics such as precision and recall.

6. User Interface Design and Integration: (a) Design wireframes and high-fidelity prototypes for the user interface. (b) Develop the front-end application using technologies like HTML, CSS, and JavaScript. (c) Integrate the AI model with the front-end to provide real-time movie recommendations

Project Requirements



7. Testing and Refinement: (a) Conduct usability testing to gather feedback. (b) Identify and fix bugs or issues in the system. (c) Refine the AI model and user interface based on user feedback.

8. Final Presentation and Documentation: (a) Prepare a presentation covering the project objectives, process, outcomes, and learnings. (b) Document the development process, challenges faced, and solutions implemented. (c) Submit the final project along with a detailed report.

Reminder - This information can also be found in Coursera's assignments

Deliverables

Project Plan Document: Detailed timeline, milestones, and assigned tasks.

User Needs Assessment Report: Summary of user movie preferences and viewing behavior.

Data Collection and Preprocessing Report: Description of data collection methods and preprocessing steps.

AI Model and Performance Report: Details of the chosen algorithm, training process, and model performance metrics.



Deliverables

Interactive User Interface: Functional front-end application with integrated AI movie recommendations.

Usability Testing Report: Feedback from users and refinements made to the system.

Final Project Presentation: Comprehensive presentation summarizing the entire project.

Project Documentation: Detailed documentation of the project development process and outcomes.



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Expectations

Weekly Time Commitment:

- Spend a minimum of 5 hours per week on the project.

Independent Learning:

- Take initiative in learning new tools and techniques as needed.
- Utilize online resources, tutorials, and documentation to support your project work.

Quality and Thoroughness:

- Ensure high quality and thoroughness in all aspects of the project.
- Pay attention to detail and strive for excellence in your deliverables.



Weekly Check-ins

Coursera Discussions:

- Participate in weekly check-ins to discuss progress and challenges.
- Provide updates on milestones and deliverables.

